

Adaptive Computation and Machine Learning

COMS 4030A and COMS 7047A

TEST 10 April 2025

Time: 90 minutes

Question 1. (7 marks)

Consider a neural network that is being trained using sum-of-squares loss function

$$L = \frac{1}{2} \sum_{\ell} (y_{\ell} - t_{\ell})^2,$$

where ℓ ranges over all output nodes.

Recall that for a node n , we set $\delta_n = \frac{\partial L}{\partial z_n} \Big|_{\mathbf{x} \mathbf{t} \mathbf{W}}$ (using the notation from the Lecture Notes).

(a) (3 marks) If n is an output node of the neural network, show that

$$\delta_n = (a_n - t_n) \left(\frac{dy_n}{dz_n} \Big|_{\mathbf{x} \mathbf{t} \mathbf{W}} \right).$$

(b) (4 marks) If n is a node in a hidden layer in the neural network, show that

$$\delta_n = \left(\sum_{\ell} \delta_{\ell} w_{n\ell} \right) \left(\frac{dy_n}{dz_n} \Big|_{\mathbf{x} \mathbf{t} \mathbf{W}} \right)$$

where the ℓ in the summation ranges over all nodes ℓ in the layer after n 's.

Question 2. (3 marks)

Explain what the cross-entropy loss function is and how it is used in training neural networks.

Question 3. (3 marks)

Explain what batch gradient descent is and describe how it works.

Question 4. (5 marks)

Use the entropy method to create a decision tree with the dataset S given below.

Show all your calculations and draw the final decision tree.

$S :$	F_1	F_2	target
	t	x	Y
	f	y	N
	t	z	Y
	t	x	N
	f	z	Y
	f	x	N
	t	y	N
	t	x	Y
	f	z	Y
	f	y	N
	f	y	Y

Question 5. (3 marks)

Compute $\text{Gain}(T, F_1)$ for the dataset T below using the mean square error (MSE) and with split point of 0.75 for attribute F_1 . Show all your calculations.

$T :$	F_1	target
	0.3	1.2
	0.7	1.6
	0.5	0.9
	0.1	0.7
	0.9	1.3
	0.8	0.9

Question 6. (4 marks)

(a) (3 marks) Describe the process of creating a random forest for a given dataset.

State the hyperparameters that are required for the process.

(b) (1 mark) Describe how a random forest is used to predict the target value of a given input.

Total marks: 25